Sign reversal of the Fano asymmetry in a coupled RLC circuit swept through the EIT configuration

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Two RLC meshes coupled by a shared capacitor reproduce electromagnetically induced transparency (EIT) when the two meshes resonate at the same frequency. This work sweeps the dark-mesh capacitor C_1 through that configuration in two damping regimes, 20 configurations in total, and records the bright-mesh current. Mapped onto the atomic Λ system, C_1 acts as a pure control-field detuning. The peak asymmetry $a = \ln(h_L/h_R)$ changes sign continuously with the detuning in both regimes, with slopes of 0.037 ± 0.006 and 0.144 ± 0.011 per kHz, a ratio of 3.9 ± 0.7 set by the damping. The model-selection test of Anisimov, Dowling and Sanders, applied with the functions of Giner *et al.*, classifies every damped lineshape as EIT-like and every undamped lineshape as Autler-Townes splitting. The zero of the asymmetry is displaced to -1.45 ± 0.29 and -1.23 ± 0.15 kHz of detuning, away from the symmetric configuration, which a model with constant unequal damping cannot reproduce. Fits with constant effective resistances show a dark-mesh loss about 87% above its DC value and a bright-mesh loss about 20% above, well beyond the skin effect.

I. INTRODUCTION

The Garrido Alzar work [1] on a classical analog of electromagnetically induced transparency showed in 2002 that two RLC meshes coupled by a capacitor reproduce EIT. A narrow transparency window inside a broad absorption line is observed at the point where the meshes resonate at the same frequency. It also shows that as one decreases the coupling capacitor value the peaks of absorption spread out. That paper stopped there. The question this paper answers is what the same circuit does everywhere else, other than this specific setup where the two meshes resonate at the same frequency.

Satpathy and colleagues [2] demonstrated Fano interference experimentally in a coupled RLC circuit. Iizawa and colleagues [3] showed that the asymmetry parameter of such a circuit is a true Fano q. None of them mapped the continuous asymmetry through zero and neither varied the damping. In atomic physics the same question, whether a transparency window is EIT or Autler-Townes splitting (ATS), has an objective test based on Akaike's information criterion, proposed by Anisimov, Dowling and Sanders [4] and tested experimentally with cold atoms by Giner *et al.* [5].

This work does three things beyond them: it measures the sign reversal of the asymmetry as a single capacitor is swept through the symmetric configuration, in two damping regimes; it shows that damping at a fixed coupling moves the same circuit between a Fano-like feature and a resolved doublet, and quantifies this with the test of Refs. [4, 5]; and it treats the deviations from the ideal picture as results, particularly referencing the displacement of the lineshape-symmetry crossover from the symmetrical configuration. To connect the circuit with the atomic case, the circuit is mapped term by term onto the atomic Λ system, so that the swept capacitor becomes the detuning of the control field.

II. PHYSICAL PICTURE

The asymmetry of the lineshape is controlled by the detuning between the two meshes' resonant frequencies and the sign of the asymmetry follows the sign of the detuning. In the ideal circuit the symmetric window is, to within a fraction of a kHz, the zero-asymmetry member of a continuous family traversed by a single capacitor. The loops' resistances set the scale for the asymmetry and the notch depths. The use of a Fano description is appropriate when the lineshape describes a narrow feature along a broad, slowly varying background. At fixed coupling, the width of the lineshapes is determined by the damping, so whether a Fano description is a good approximation of the lineshape depends on whether the response is a narrow feature on a wide background or a resolved doublet.

The bright mesh is the one that "sees" the generator or the laser in the EIT, and its impedance is its resistance plus a reactance[7] that is zero at its resonance, negative below it and positive above it. The current is largest where the total impedance is lowest, which is the mechanism that allows for the peaked lineshapes to occur. Because C_c is shared with the dark mesh, its presence adds impedance to what the source sees. The extra impedance changes based on where you are relative to the dark mesh's resonance, being below it will mean a positive reactance and above it mean a negative reactance. So the sign changes across the dark resonance and is pure resistance at the centre. At the dark resonance the large added resistance makes the total impedance large, which means the bright mesh current drops. This is the transparency window dip seen. On either side of the notch, the total reactance is either small because the bright mesh's reactance is in the opposite sign as the dark mesh's extra added reactance meaning they either cancel out, giving a small total impedance, or it remains high because the dark mesh's extra reactance is toward the same direction as the bright mesh's reactance so they add, leaving

the current low and producing a slowly upward curving line-shape.

If the bright mesh resonates at the same frequency as the dark mesh, its own reactance is zero at the notch so the cancellation is close to equal and opposite and the window is close to symmetric. The notch sits at the dark resonance so if the bright mesh resonates below the dark resonance, meaning a smaller C_1 so a higher dark mesh resonance frequency ω, the inductive term dominates the bright mesh's reactance at the notch, meaning its reactance there is positive. For the dark mesh, below the dark resonance the reactance is dominated by the capacitive term so it is negative, but the reactance it returns to the bright mesh has the opposite sign, so the extra reactance added is positive. On this side near the dark resonance the total impedance stays high and the current stays low. Above the dark resonance the reactance added to the total impedance is dominated by the inductive term so what it returns to the bright mesh is negative. It cancels the bright mesh's positive reactance close to the notch, giving a peak on the high frequency side. But this close to the dark resonance the dark mesh also returns a large resistance so that peak is always short. Further below the notch two things happen at once. The dark mesh moves away from its resonance so both the reactance and resistance returned shrink toward zero. And the bright mesh passes below its own resonance so the reactance becomes dominated by the capacitive term and negative. So, far below the dark resonance frequency the bright mesh's small negative reactance and the small positive reactance returned by the dark mesh cancel a second time, and because the dark mesh is now far away from its resonance it adds almost no resistance, so the total impedance is close to the bright mesh's resistance alone, which is also where the loop resistors enter the peak asymmetry. This interaction between the reactances when the dark mesh's resonance is higher than the bright mesh's results in a higher peak on the low-frequency side and a lower peak on the high-frequency side. This all reverses if C_1 is larger than the symmetric value, meaning the dark resonance would be below the bright resonance, ultimately resulting in a higher peak on the high-frequency side and a lower peak on the low-frequency side. The returned resistance is proportional to the dark mesh resistance R_1. With no resistance in the dark mesh it vanishes and both peaks reach the same height, so the asymmetry exists only because the dark mesh dissipates.

A. Correspondence with the atomic Λ system

In the atomic system a weak probe field drives the transition $|g\rangle \to |e\rangle$ and a control field of Rabi frequency Ω couples $|e\rangle$ to a second ground state $|s\rangle$. For a control field detuned by Δ_c from its transition, the linear susceptibility seen by the probe at detuning δ is [5, 6]

$$\chi(\delta) \propto -\frac{\delta_2 + i\gamma_{gs}}{(\delta + i\gamma_{eg})(\delta_2 + i\gamma_{gs}) - \Omega^2/4}, \qquad \delta_2 = \delta - \Delta_c, \tag{1}$$

where γ_{eg} is the decay rate of the optical coherence, γ_{gs} the dephasing rate of the ground-state coherence, and the absorption is proportional to $\mathrm{Im}\,\chi$. Eq. (1) of Ref. [5] is the case $\Delta_c = 0$.

For the circuit the impedance seen by the source is $Z_{\rm in} = Z_2 + 1/[(\omega C_c)^2 Z_1]$, with Z_1 and Z_2 the impedances of the dark and bright meshes. Near resonance each mesh impedance is $Z_k \approx R_k + j\,2L_k(\omega - \omega_k)$, so the admittance is

$$Y(\omega) = \frac{1}{Z_{\rm in}} \approx \frac{-j}{2L_2}\,\frac{\delta_2 - j\gamma_1}{(\delta - j\gamma_2)(\delta_2 - j\gamma_1) - \Omega^2/4}, \tag{2}$$

with $\delta = \omega - \omega_2$, $\delta_2 = \omega - \omega_1$, $\gamma_k = R_k/2L_k$ and

$$\Omega = \frac{1}{\omega C_c \sqrt{L_1 L_2}}. \tag{3}$$

Eq. (2) has the same form as Eq. (1), with $j \to -i$ the only change, which comes from the engineering sign convention. The power taken from the source per $|V_s|^2$ is $\mathrm{Re}\,Y$, which plays the role of the absorption. The full correspondence is given in Table I.

Ω does not contain C_1, so C_1 only moves ω_1 and acts as a pure control detuning, $\Delta_c = \omega_1 - \omega_2$. Physically sweeping C_1 moves the dark mesh resonance through Eq. (5); in the atomic language it detunes the control laser. The atomic theory then predicts three things for the circuit. First, the transparency sits at two-photon resonance, $\delta_2 = 0$, so $f_T = f_1$. Second, the two peaks are dressed states that repel as Δ_c passes through zero instead of crossing. For the lossless circuit the condition $Z_1 Z_2 = -1/(\omega C_c)^2$ gives $L_1 L_2(\omega^2 - \omega_1^2)(\omega^2 - \omega_2^2) = 1/C_c^2$, so the dressed-state frequencies are exactly

$$f_\pm^2 = \frac{f_1^2 + f_2^2}{2} \pm \frac{1}{2}\sqrt{\left(f_1^2 - f_2^2\right)^2 + 4\kappa^4}, \tag{4}$$

with $\kappa^4 = 1/[(2\pi)^4 L_1 L_2 C_c^2]$, which reduces to the atomic $\delta_\pm = \Delta_c/2 \pm \frac{1}{2}\sqrt{\Delta_c^2 + \Omega^2}$ when the coupling is weak. The coupling here is not weak, Ω being half of f_2, so Eq. (4) is used. Third, for $\Delta_c \neq 0$ the two dressed states are unequal mixtures of the two meshes. The one that is mostly bright is the "atom-like" state, broad and tall, and the one that is mostly dark is the "Raman-like" state, narrow and short. Which side each one sits on is set by the sign of Δ_c, which is the sign of a. The difference in their heights requires $\gamma_{gs} \neq 0$, which is the atomic version of the statement that the asymmetry exists only because the dark mesh dissipates.

Table II gives the atomic parameters reached by each C_1. The same sweep covers the same range of Δ_c/Ω in both regimes, -0.31 to $+0.51$, because the coupling never changes. In units of the optical linewidth it does not: the damped circuit stays within Δ_c/γ_{eg} of -0.26 to $+0.56$ and the undamped one reaches -1.21 to $+2.01$, about 3.6 times further from resonance.

TABLE I. Correspondence between the atomic Λ system and the circuit. Rates and frequencies are given divided by 2π. Damped values use the fitted resistances of Sec. III G; the undamped γ_{eg} uses a representative 37 Ω and is indicative only.

Atomic	Circuit	Value
Probe detuning δ	$\omega - \omega_2$ (generator)	swept 3 to 60 kHz
Probe transition $\|g\rangle \to \|e\rangle$	Bright mesh	$f_2 = 21.46$ kHz
Ground-state coherence $\|g\rangle$–$\|s\rangle$	Dark mesh	$f_1 = 18.15$ to 26.95 kHz
Control Rabi frequency Ω	$1/(\omega C_c\sqrt{L_1 L_2})$	10.7 kHz
Control detuning Δ_c	$\omega_1 - \omega_2$, set by C_1	−3.3 to +5.5 kHz
Optical decay γ_{eg}	$R_2/2L_2$	9.8 kHz (damped), 2.7 kHz (undamped)
Ground-state dephasing γ_{gs}	$R_1/2L_1$	1.4 kHz
Absorption $\mathrm{Im}\,\chi$	$\mathrm{Re}\,(1/Z_{\mathrm{in}})$	not recorded

TABLE II. Control detuning reached by each C_1. f_1 from Eq. (5), $\Delta_c/2\pi = f_1 - f_2$ with $f_2 = 21.46$ kHz, $\Omega/2\pi = 10.7$ kHz, and $\gamma_{eg}/2\pi = 9.8$ kHz (damped) and 2.73 kHz (undamped, representative 37 Ω). A dash marks a value of C_1 not measured in that regime.

C_1 (nF)	f_1 (kHz)	$\Delta_c/2\pi$ (kHz)	Δ_c/Ω	Δ_c/γ_{eg} damped	Δ_c/γ_{eg} undamped
48.43	26.95	+5.48	+0.51	+0.56	+2.01
70.27	23.93	+2.47	+0.23	+0.25	+0.90
79.88	23.06	+1.60	+0.15	+0.16	–
92.03	22.19	+0.73	+0.07	+0.07	–
96.11	21.94	+0.48	+0.04	+0.05	–
100.9	21.67	+0.21	+0.02	+0.02	+0.08
109.5	21.23	−0.23	−0.02	−0.02	–
112.91	21.08	−0.38	−0.04	–	−0.14
117.87	20.87	−0.59	−0.06	−0.06	–
123.5	20.64	−0.82	−0.08	−0.08	–
128.61	20.46	−1.00	−0.09	−0.10	–
139.71	20.09	−1.37	−0.13	−0.14	−0.50
153.17	19.72	−1.75	−0.16	−0.18	–
193.4	18.87	−2.59	−0.24	−0.26	−0.95
246.71	18.15	−3.31	−0.31	–	−1.21

III. EXPERIMENTAL METHOD

A. Circuit

The circuit consists of two RLC meshes coupled by a shared capacitor $C_c = 102.7$ nF, following the topology of Garrido Alzar *et al.* [1], shown in Fig. 1. The driven (bright) mesh contains inductor $L_2 = 1.08$ mH, capacitor $C_2 = 101$ nF, a series resistor R_2 and the winding resistance of L_2 at 9.5 Ω. It is driven by the Instrumentherm GF-550 function generator that has an output impedance of 50 Ω, which supplies a sinusoidal voltage of the maximum amplitude in the undamped runs and a constant lower amplitude for the damped runs. The undriven (dark) mesh contains $L_1 = 1.06$ mH and a variable capacitor C_1; its only resistance is the 10.0 Ω DC winding resistance of L_1, and no resistor was added. Two damping regimes were used, differing only in the bright mesh. The damped runs used $R_2 = 50.9$ Ω and the nominal output impedance of 50 Ω. The undamped runs used $R_2 = 6.2$ Ω and a 3.16 Ω shunt added in parallel to the 50 Ω output resistance to decrease it to 2.97 Ω. The total bright mesh series resistance at DC was therefore 110.4 Ω in the damped regime and 18.7 Ω in the undamped regime. Because C_c is shared the dark mesh resonant frequency is set by L_1 in series with C_1 and C_c,

$$f_1 = \frac{1}{2\pi}\sqrt{\frac{1}{L_1}\left(\frac{1}{C_1} + \frac{1}{C_c}\right)}, \tag{5}$$

so sweeping C_1 alone changes the detuning between the two meshes while leaving the bright mesh and coupling unchanged.

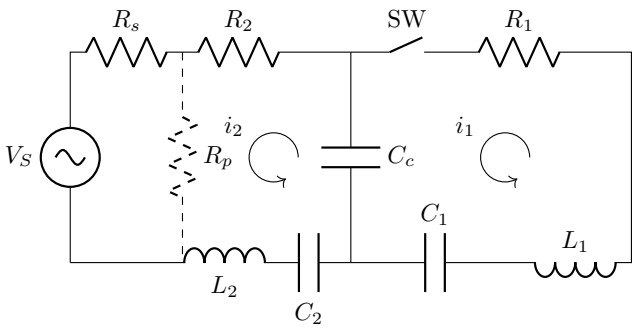


FIG. 1.

The component values are listed in Table III. Inductances were read from a lab-built LCR meter of unknown test frequency. Capacitances and resistances were measured using the same LCR meter.

TABLE III. Circuit components.

Component	Value	Method
L_2 (bright)	1.08 mH	Lab-built LCR meter
L_1 (dark)	1.06 mH	Lab-built LCR meter
R_{L2}	9.5 Ω	Lab-built LCR meter
R_{L1}	10.0 Ω	Lab-built LCR meter
C_2	101 nF	Lab-built LCR meter
C_c	102.7 nF	Lab-built LCR meter
C_1	varied (Tables IV, V)	Lab-built LCR meter
R_2, damped	50.9 Ω	Lab-built LCR meter
R_2, undamped	6.2 Ω	Lab-built LCR meter
R_1	0 Ω	No resistor present
Source R	50 Ω	Nominal
Source R, undamped	2.97 Ω	50 Ω ∥ 3.16 Ω shunt

Two damping regimes were measured, defined by the values of R_2 and the source resistance in Table III; all other components were kept the same between regimes. All the capacitances for each run were built from bank capacitors in parallel. The symmetric configuration can be defined as $\omega_1 = \omega_2$, which for the measured inductances $L_1 = 1.06$ mH and $L_2 = 1.08$ mH means that it is not at $C_1 = C_2$, rather it is at $C_1 = 104.9$ nF. The nearest measured configuration is 100.9 nF in both regimes. One run was excluded, for the use of an unidentified bank capacitor.

B. Measurement procedure

For each configuration, the voltage across L_2 was recorded as a function of the generator frequency. Since $V_{L2} = \omega L_2 I_2$, the bright mesh current is proportional to V_{L2}/f, which is the quantity reported, in arbitrary units. This is what is measured in Ref. [1], obtained there through integration and here by division.

TABLE IV. Damped regime configurations.

Run sheet	Capacitors in parallel (nF)	Total capacitance (nF)
1	48.43	48.43
3	48.43 // 21.84	70.27
4	48.43 // 27.39 // 4.06	79.88
5	48.43 // 21.84 // 21.76	92.03
6	96.11	96.11
7	96.11 // 4.8	100.9
8	96.11 // 8.6 // 4.8	109.5
9	96.11 // 21.76	117.87
10	96.11 // 27.39	123.5
11	96.11 // 24.3 // 8.2	128.61
12	96.11 // 21.84 // 21.76	139.71
13	96.11 // 48.86 // 8.2	153.17
14	96.11 // 48.86 // 48.43	193.4

TABLE V. Undamped regime configurations.

Run sheet	Capacitors in parallel (nF)	Total capacitance (nF)
15	48.43	48.43
16	48.43 // 21.84	70.27
17	96.11 // 4.8	100.9
18	96.11 // 8.6 // 8.2	112.91
19	96.11 // 21.84 // 21.76	139.71
20	96.11 // 48.86 // 48.43	193.4
21	150.6 // 96.11	246.71

The voltage was read from channel 1 of a Tektronix TDS 1001B two-channel oscilloscope of 40 MHz and 500 MS/s with a 10× probe across L_2, using the instrument's automatic peak-to-peak measurement. The vertical scale was adjusted as necessary so the trace fit the screen. Three readings were taken at each frequency and averaged. The waveform was checked to be sinusoidal before each reading and the amplitude was set once at the start of each regime and unchanged throughout.

The frequency was swept from 3 to 60 kHz in the damped regime and 7 to 60 kHz in the undamped regime. Steps of 1 kHz were used far from the predicted notch frequency f_1 of Eq. (5) and reduced to 0.5 kHz within 8 kHz of it, and decreased to 0.25 kHz within 4 kHz, on both sides. Above 45 kHz, the steps were increased to 5 kHz because the response is featureless and was only captured to observe the trend. In the undamped regime the measurements below 7 kHz were excluded because their waveforms were deformed and the signal was under 15 mV.

C. Reference lineshape

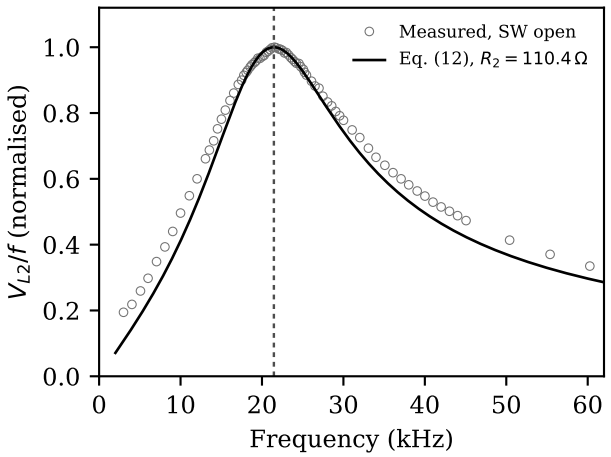


FIG. 2.

Illustrated by Fig. 2 the reference line-shape is the

set of results collected when the switch SW is opened, meaning the dark mesh is disconnected and the bright mesh is driven alone and recorded. The measurements were taken in the damped regime and it was taken to serve as the baseline against which the notch and the Fano feature are defined and it is the pump off curve of Ref. [1], Eq. (12). The predicted $f_2 = 21.46$ kHz from the same expression as Eq. (5) with L_2, C_2, C_c. The measured $f_2 = 21.53 \pm 0.25$ kHz, within 0.3%.

D. Derived quantities

From each line-shape five quantities were extracted. The first, peak frequencies f_L and f_R are the frequencies of the largest measured V_{L2}/f below and above f_1, respectively. The second, the transparency window frequency f_T is that of the smallest measured V_{L2}/f between the two peaks. Third, the transparency offset is calculated by $(f_T - f_1)/f_1$ with f_1 from Eq. (5). Fourth, the peak ratio h_L/h_R is the ratio of V_{L2}/f at f_L to that at f_R, and the peak asymmetry a is defined by $a = \ln(h_L/h_R)$, which is zero for the symmetric peaks and changes sign when the asymmetry is reversed. Fifth, the transparency depth T_d is V_{L2}/f at f_T divided by the higher of the two peaks. Table VI details the derived quantities for all configurations.

E. Uncertainties

All frequencies reported are measurement-grid points so their uncertainty is taken as half the local step. This gives 0.07 to 0.13 kHz for f_T, 0.10 to 0.27 kHz for f_L and f_R, and about 0.5 kHz for f_L in the two 48.43 nF configurations where the peak was still in the 1 kHz steps. The generator frequency was taken as exact.

The uncertainty of V_{L2}/f is the largest one between: half the range of the three values recorded and the oscilloscope's reading resolution of 1%. Since the values frequently agreed, the resolution sets the floor. Any common error gain of probe and scope cancels out in ratios of readings taken at the same vertical scale. Ratios of readings taken at different vertical scales, as for T_d, also carry the oscilloscope's vertical gain accuracy between scales, which is not included.

Propagating the errors forward it is observed that with the 1% floor the h_L/h_R carries $\pm$ 0.01 to 0.04, the transparency offset carries an uncertainty of 0.3 to 0.6 percentage points, and T_d of $\pm$ 0.002 to 0.006. Per-run values are stated in Table VII.

These uncertainties do not include the reassembly of the capacitor bank between runs. One configuration, 112.91 nF in the undamped regime, was re-measured around the peaks and the window in a separate session. It gave $a = 0.18$ and $T_d = 0.115$ against 0.12 and 0.159 in the original run, so the run-to-run reproducibility is larger than the propagated uncertainties.

F. Model

The line-shapes are compared with the exact two mesh model of Ref. [1], Eqs. (9) to (11). The quantity compared is the current amplitude, $|I_2| = A/\sqrt{{p_1}^2 + {p_2}^2}$ with p_1 and p_2 as defined there. It is not the absorbed power of their Eq. (9) for three reasons. First, the two quantities have different shapes where it is important. The power is $P_2 = |I_2|^2 p_1$ and $p_1 = R_2 + R_1/[(\omega C_c)^2 |Z_1|^2]$ contains resistance passed into the bright mesh from the dark mesh. At f_1 this value is largest where $|Z_1|$ is smallest, so across the transparency window P_2 falls less steeply than $|I_2|^2$ and the window in power is less deep and somewhat displaced relative to the window in current. This effect is negligible far away from f_1 where $P_2 \propto |I_2|^2$. So, any value derived from the window such as its depth and position compared to either peak takes a different value based off of what is plotted. The peak height ratios if using power would only be the square of those using current. Second, what was measured was the current. The probe reads $V_{L2} = \omega L_2 |I_2|$ so $V_{L2}/f \propto |I_2|$ as given in Sec. III B. Third, the power cannot be obtained from the data because $P_2 = |V_s||I_2| \cos\phi$ which requires the source voltage and the phase between it and the current at every frequency, none of which was recorded. So, the only route to P_2 from V_{L2}/f is to multiply p_1 taken from the model which would just compare the model against itself. Ref. [1] also measured current and treated it as sharing the frequency dependence of P_2, which holds when R_1 is small. So, using current, the curves in Figs. 2, 3 and 4 are the expression of this evaluated at the component DC values of Table III. Only the overall scale A was adjusted using a least squares regression.

G. Fits with constant effective resistances

The DC model leaves the widths and depths of the measured lineshapes unexplained, so a second comparison was made with the resistances floated. Here R_1 and R_2 stand for the total series resistance of each mesh, including the windings and, for the bright mesh, the source. Both were fitted as constants per run together with the scale A, starting from the DC values, by unweighted least squares on the normalised V_{L2}/f over the full range.

In the damped regime the fit is stable across all 13 runs and gives $R_1 = 18.7 \pm 1.1\ \Omega$ and $R_2 = 133 \pm 4\ \Omega$, against 10.0 and 110.4 Ω at DC. The misfit drops from 6 to 8% of the peak to 2 to 4%. The reference run alone gives $R_2 = 132.7 \pm 1.3\ \Omega$, which agrees. In the undamped regime the misfit drops from 10 to 14% to 6 to 9%, but the values do not settle, R_1 ranging from 28 to 10 Ω and R_2 from 20 to 51 Ω across runs. A constant effective resistance therefore describes the damped regime but not the undamped one. A frequency-dependent resistance was not fitted, which is out of the scope of this work.

TABLE VI. Derived quantities for all configurations. Frequencies in kHz. f_1 is the predicted dark-mesh resonance of Eq. (5); f_L, f_R and f_T are the measured peak and transparency frequencies; h_L/h_R is the peak height ratio, $a = \ln(h_L/h_R)$ the peak asymmetry, and T_d the transparency depth, all in V_{L2}/f. Uncertainties are given in Sec. III E.

Regime	C_1 (nF)	f_1	f_L	f_R	f_T	$(f_T - f_1)/f_1$ (%)	h_L/h_R	a	T_d
Damped	48.43	26.95	18.01	30.77	27.08	+0.5	1.26	+0.233	0.392
Damped	70.27	23.93	16.09	29.02	23.74	−0.8	1.16	+0.148	0.370
Damped	79.88	23.06	15.49	28.54	22.75	−1.4	1.13	+0.120	0.354
Damped	92.03	22.19	15.04	27.53	21.75	−2.0	1.10	+0.100	0.354
Damped	96.11	21.94	14.49	27.55	21.52	−1.9	1.05	+0.051	0.374
Damped	100.9	21.67	14.53	27.55	21.25	−1.9	1.02	+0.025	0.369
Damped	109.5	21.23	14.55	27.05	20.76	−2.2	1.09	+0.090	0.331
Damped	117.87	20.87	14.00	27.07	20.55	−1.5	1.04	+0.037	0.336
Damped	123.5	20.64	14.03	26.54	20.26	−1.9	1.03	+0.028	0.329
Damped	128.61	20.46	13.52	26.51	20.08	−1.8	0.98	−0.023	0.320
Damped	139.71	20.09	13.08	26.51	19.76	−1.7	1.03	+0.031	0.319
Damped	153.17	19.72	13.01	26.00	19.24	−2.4	0.96	−0.041	0.313
Damped	193.4	18.87	12.06	25.50	18.51	−1.9	0.98	−0.024	0.285
Undamped	48.43	26.95	18.07	30.75	27.51	+2.1	2.47	+0.905	0.134
Undamped	70.27	23.93	17.07	29.05	24.52	+2.5	1.75	+0.562	0.166
Undamped	100.9	21.67	15.55	27.52	22.08	+1.9	1.18	+0.166	0.172
Undamped	112.91	21.08	15.01	27.04	21.25	+0.8	1.12	+0.117	0.159
Undamped	139.71	20.09	14.06	26.53	20.52	+2.1	0.95	−0.053	0.172
Undamped	193.4	18.87	13.05	26.50	19.24	+1.9	0.86	−0.151	0.111
Undamped	246.71	18.15	12.04	26.00	19.07	+5.0	0.84	−0.171	0.126

TABLE VII. Uncertainties in the derived quantities of Table VI. Frequency uncertainties are half the local grid step, in kHz. Offset uncertainty is $\delta f_T/f_1$, in percentage points. Uncertainties in h_L/h_R and T_d are absolute, propagated in quadrature from the height uncertainty at each point, taken as the larger of 1% of the reading and half the range of the three readings.

Regime	C_1 (nF)	δf_L	δf_R	δf_T	δOffset (%)	$\delta(h_L/h_R)$	δT_d
Damped	48.43	0.48	0.12	0.07	0.3	0.021	0.006
Damped	70.27	0.21	0.25	0.10	0.4	0.016	0.005
Damped	79.88	0.24	0.26	0.10	0.5	0.016	0.005
Damped	92.03	0.24	0.23	0.13	0.6	0.016	0.005
Damped	96.11	0.20	0.24	0.11	0.5	0.015	0.005
Damped	100.9	0.24	0.27	0.12	0.6	0.016	0.005
Damped	109.5	0.27	0.23	0.12	0.5	0.015	0.005
Damped	117.87	0.25	0.24	0.10	0.5	0.015	0.005
Damped	123.5	0.24	0.26	0.12	0.6	0.015	0.005
Damped	128.61	0.22	0.23	0.09	0.4	0.014	0.005
Damped	139.71	0.21	0.22	0.12	0.6	0.015	0.005
Damped	153.17	0.25	0.23	0.10	0.5	0.014	0.004
Damped	193.4	0.23	0.24	0.12	0.6	0.014	0.004
Undamped	48.43	0.48	0.10	0.12	0.4	0.035	0.002
Undamped	70.27	0.23	0.21	0.12	0.5	0.025	0.002
Undamped	100.9	0.24	0.26	0.07	0.3	0.017	0.002
Undamped	112.91	0.24	0.24	0.10	0.5	0.016	0.002
Undamped	139.71	0.23	0.24	0.11	0.5	0.013	0.002
Undamped	193.4	0.22	0.22	0.12	0.6	0.015	0.002
Undamped	246.71	0.23	0.23	0.09	0.5	0.012	0.002

IV. RESULTS

Figure 2 shows the bright mesh driven without the dark mesh connected, in the damped regime. This is the background of every other lineshape in the damped regime and the pump-off curve of Ref. [1] that follows Eq. (12). The measured peak is at 21.53 ± 0.25 kHz against the predicted $f_2 = 21.46$ kHz, a 0.3% agreement that confirms L_2, C_2 and C_c at the operating frequency. The measured line-shape is broader than the DC value 110.4 Ω on both sides of the peak. The same expression using 133 Ω instead reproduces the full curve (Sec. III G). Meaning, the bright mesh dissipates around 20% more at 20 kHz than its DC resistance accounts for. This excess is the first measurement of the effective loop resistance,

and it reappears in every coupled line-shape below.

Figure 3 shows the damped regime for all 13 values of C_1. Every graph has the same structure, a broad background peaked near the bright resonance, a transparency window at f_T, and two peaks whose relative heights change with C_1. With no parameters adjusted, the DC model predicts the position of the window to within 0.45 kHz of the measured f_T and both peaks to within 0.85 kHz in every run with the measured peaks slightly farther apart than the model's. As C_1 increases the frequency of the window decreases, following f_1 of Eq. (5) from 26.95 to 18.87 kHz, while the peak height ratio h_L/h_R falls from 1.26 to 0.98 (Table VI). The measured window is shallower than the DC model in every sweep, $T_d = 0.29$ to 0.39 against model values between 0.15 and 0.25, and the ratio grows with C_1 from 1.6 to 1.9. Also, the measured f_T lies 1.4 to 2.4% below the predicted f_1 in 11 of the 13 runs.

Figure 4 shows the undamped regime for 7 values of C_1. With the bright mesh resistance reduced from about 110.4 to 18.7 Ω the same circuit gives two resolved peaks separated by a deep window of $T_d = 0.11$ to 0.17, a resolved doublet. The peak height ratio ranges from $h_L/h_R = 2.47$ to 0.84 as C_1 increases. The model using the DC values also reproduces the left peak to within 0.4 kHz, but the measured right peak sits 0.7 to 1.4 kHz above the model's and is lower and broader than the model and the tail above 30 kHz is above it in every value of C_1. The window is 2 to 4 times shallower than the model predicts. The measured f_T lies from 1.9 to 2.5% above the predicted f_1 values in 5 of 7 runs. The two regimes therefore exemplify two aspects of this family of circuits, as the C_1 is varied the asymmetry of the two peaks shifts and the damping determines whether it is a broad background or a resolved doublet.

Figure 5 shows the peak asymmetry $a = \ln(h_L/h_R)$ from Table VI against the detuning $f_1 - f_2$ between the dark and bright resonances. In both regimes a rises and crosses 0, the taller peak moves from the high frequency side at the negative detuning to the low frequency side at the positive detuning. The undamped asymmetry spans −0.17 to +0.9 while the damped ranges −0.04 to +0.23 over the same detuning range of −3.3 to +5.5 kHz. The same reversal happens in both regimes just about four times smaller in the damped regime. In the undamped regime a changes sign between −0.4 and −1.4 kHz. In the damped regime it changes sign between −0.8 and −1.8 kHz. In both regimes the sign change occurs in the negative detuning. Figure 6 shows the offset of the transparency window against detuning. The two regimes create two separate populations that have no dependence on detuning. The damped f_T lies 1.4 to 2.4% below the predicted f_1 in 11 of the 13 runs and the undamped f_T lies 1.9 to 2.5% above it for 5 of the 7 runs. The exceptions were the two largest positive values of detuning in the damped regime, and the most negative value of detuning (+5.0%) and the 112.91 nF run (+0.8%) in the undamped regime.

Figure 7 shows the transparency depth against detuning. The damped window is shallow, T_d rising from 0.29 at the most negative to 0.39 at the most positive value of detuning. Comparatively, the undamped window depth T_d is relatively deep varying between 0.11 and 0.17 with no discernible trend. A sixfold change to the value of the bright mesh resistance equates to an increase of factor roughly 2 to 3 of the depth of the transparency window. Curiously, the relation to the detuning appears only in the damped regime.

V. ANALYSIS

A. One continuous family

Linear fits of a against detuning over $|f_1 - f_2| < 2.6$ kHz give slopes of 0.037 ± 0.006 per kHz in the damped regime and 0.144 ± 0.011 per kHz in the undamped regime, a ratio of 3.9 ± 0.7. The zero crossings are at -1.45 ± 0.29 kHz (damped) and -1.23 ± 0.15 kHz (undamped), which agree. The damped scatter about the line is 0.026, about 1.7 times the per-point δa of 0.015, which is consistent with the run-to-run reproducibility of Sec. III E. The sign of a follows the detuning as the physical picture predicts. The 100.9 nF run sits on the line with its neighbours, so the symmetric window is one member of the family, not a special state. The 109.5 nF run sits 2.9σ off the line, possibly from the reassembly of the capacitor bank.

The same continuity appears in the atomic language of Sec. II A. Figure 8 shows the peak and window frequencies against f_1, that is against the control detuning. The undamped peaks follow the two dressed-state branches of Eq. (4), computed from the component values with nothing fitted, and repel instead of crossing as f_1 passes f_2. The lower branch agrees with the measured f_L to within 0.45 kHz in every run and the upper branch lies 0.8 to 1.6 kHz below the measured f_R, the same right-peak deviation seen in Fig. 4. The windows of both regimes sit on the diagonal $f_T = f_1$ to within 2.5% in 19 of 20 runs, which is the atomic statement that transparency occurs at two-photon resonance.

B. Damping sets the scale and the character

The slope ratio of 3.9 is smaller than the ratio of the DC bright mesh resistances, 5.9. With the constant fits of Sec. III G it is $133/37 = 3.6$, taking 37 Ω as a representative undamped value, although the undamped fitted values are unstable. In the atomic language the slope ratio is the ratio of how far from resonance the same C_1 sweep takes each regime in units of γ_{eg} (Table II).

The widths $R/(2\pi L)$ set the character of the lineshape. The dark mesh width is about 2.8 kHz. The bright mesh width is about 19.6 kHz in the damped regime and about 5.5 kHz in the undamped regime, while the peak splitting is 12 to 14 kHz in both. In the damped regime the bright

width is about 7 times the dark width and larger than the splitting, so the response is a narrow feature on a broad background. In the undamped regime the two widths are within a factor of 2 and both narrower than the splitting, so the response is a resolved doublet. More damping therefore gives less asymmetry but more Fano character.

The depth follows the dark mesh resistance. At 21 kHz the dark mesh returns about 545 Ω to the bright mesh with $R_1 = 10\ \Omega$ against 290 Ω with $R_1 = 18.7\ \Omega$, so a larger R_1 gives a shallower window. The constant-fit model gives $T_d = 0.27$ to 0.43 against the measured 0.29 to 0.39, trend with detuning included, while the DC model gives 0.15 to 0.25.

The character of each lineshape was quantified with the model-selection test of Refs. [4, 5]. Each lineshape was fitted with the two functions of Ref. [5], the EIT model of a broad Lorentzian minus a narrow one,

$$A_{\mathrm{EIT}} = \frac{\mathcal{C}_+}{1+\frac{(f-f_0-\epsilon)^2}{w_+^2/4}} - \frac{\mathcal{C}_-}{1+\frac{(f-f_0)^2}{w_-^2/4}}, \tag{6}$$

and the ATS model of two independent Lorentzians,

$$A_{\mathrm{ATS}} = \frac{\mathcal{B}_a}{1+\frac{(f-f_a)^2}{w_a^2/4}} + \frac{\mathcal{B}_b}{1+\frac{(f-f_b)^2}{w_b^2/4}}. \tag{7}$$

The window centre f_0 was left free, so both models have six parameters. The fitted quantity was $|I_2|^2 \propto (V_{L2}/f)^2$, normalised, over the region from half a peak separation below f_L to half a peak separation above f_R. For each model the Akaike information criterion is $I_j = N\ln(\mathrm{RSS}_j/N) + 2k$, with N the number of points, RSS_j the residual sum of squares and $k = 7$ including the noise variance. The Akaike weight of the EIT model is $w_{\mathrm{EIT}} = e^{-I_{\mathrm{EIT}}/2}/(e^{-I_{\mathrm{EIT}}/2} + e^{-I_{\mathrm{ATS}}/2})$, and the per-point weight $\bar{w}_{\mathrm{EIT}}$ is the same with I_j divided by N [5].

Left free, the EIT fit to the damped data finds two Lorentzians of almost the same width (13.0 and 12.9 kHz) and amplitudes of about 340 on data normalised to 1, whose difference reproduces the profile without a narrow window being carved out of a broad line. Ref. [5] meets the same solution in its strong-coupling case. To exclude it, the narrow width was limited to half the broad width and the amplitudes to 10.

Table VIII gives the result. Every damped lineshape prefers the EIT model, with $\Delta I = I_{\mathrm{ATS}} - I_{\mathrm{EIT}}$ of +37 to +113, and every undamped lineshape prefers the ATS model, with ΔI of −91 to −130. The Akaike weights are 1 and 0 in every case, because the misfit of both models is much larger than the 1% noise, so the per-point weights carry the information. They are 0.58 to 0.73 for the EIT model in the damped regime and 0.24 to 0.31 in the undamped regime (Fig. 9), close to the saturation values of 0.7 and 0.3 seen in Ref. [5]. In the damped regime the EIT preference is largest near zero detuning and falls toward negative detuning, from 0.73 to 0.58. Ref. [5] kept the control on resonance, so this dependence on Δ_c has no atomic counterpart to compare with.

TABLE VIII. Akaike test of Eqs. (6) and (7) for every configuration. $\Delta I = I_{\mathrm{ATS}} - I_{\mathrm{EIT}}$, positive when the EIT model is preferred; $\bar{w}_{\mathrm{EIT}}$ is the per-point weight of the EIT model, 0.5 meaning no preference.

Regime	C_1 (nF)	$f_1 - f_2$ (kHz)	ΔI	$\bar{w}_{\mathrm{EIT}}$
Damped	48.43	+5.48	+73	0.65
Damped	70.27	+2.47	+110	0.72
Damped	79.88	+1.60	+113	0.72
Damped	92.03	+0.73	+100	0.71
Damped	96.11	+0.48	+92	0.69
Damped	100.9	+0.21	+113	0.73
Damped	109.5	−0.23	+90	0.69
Damped	117.87	−0.59	+76	0.66
Damped	123.5	−0.82	+81	0.66
Damped	128.61	−1.00	+75	0.66
Damped	139.71	−1.37	+57	0.62
Damped	153.17	−1.75	+54	0.61
Damped	193.4	−2.59	+37	0.58
Undamped	48.43	+5.48	−107	0.28
Undamped	70.27	+2.47	−130	0.24
Undamped	100.9	+0.21	−105	0.28
Undamped	112.91	−0.38	−108	0.28
Undamped	139.71	−1.37	−99	0.30
Undamped	193.4	−2.59	−106	0.29
Undamped	246.71	−3.31	−91	0.31

The verdict does not depend on the choices made. Fitting $|I_2|$ instead of $|I_2|^2$, or a region of 0.3 or 0.8 peak separations beyond each peak instead of 0.5, gives the same verdict for every run. Because $\mathrm{Re}\,Y$ was not recorded, the test was also run on the model of Sec. III F with the fitted resistances, for three runs in each regime, once on $\mathrm{Re}\,Y$ and once on $|I_2|^2$. Both give the same verdict, but $|I_2|^2$ leans toward the EIT model: the per-point EIT weights are 0.54 to 0.67 on $\mathrm{Re}\,Y$ against 0.63 to 0.76 on $|I_2|^2$ in the damped regime, and 0.10 against 0.25 to 0.30 in the undamped regime. The undamped verdict is therefore conservative and the damped one is somewhat generous. In 12 of the 13 damped fits the broad width comes out at 17.8 to 19.9 kHz, close to the 19.6 kHz bright mesh width of the reference run, which the fit never saw, and the narrow width sits at the imposed limit, 9 to 10 kHz; the exception is the 48.43 nF run, the most detuned. This is a ratio of about 2 between broad and narrow, against about 24 in the EIT case of Ref. [5]. The fitted offset ϵ has the sign of $-\Delta_c$, as the correspondence requires, in every run with $|\Delta_c|/2\pi > 0.5$ kHz, but reaches only 17 to 48% of its size, −2.6 kHz at $\Delta_c/2\pi = +5.5$ kHz; closer to resonance it is within 0.2 kHz of zero. Eq. (6) assumes a resonant control; with $\Delta_c \neq 0$ the exact decomposition of Eq. (1) has complex amplitudes, which add a dispersive part to each Lorentzian, and a Lorentzian with a dispersive part is a Fano profile.

The damped window is therefore EIT-like but not narrow. In Giner's units [5] the damped circuit sits at $\Omega/\Gamma \approx 10.7/19.6 = 0.55$ and the undamped one at $10.7/5.5 \approx 2.0$, where $\Gamma = 2\gamma_{eg}$ is the bright mesh width. The two normal modes separate in frequency when Ω ex-

ceeds $\gamma_{eg}-\gamma_{gs}$ [5], which is about 8.4 kHz, or $\Omega/\Gamma \approx 0.43$, for the damped circuit, and the Akaike test switches to the ATS model at $\Omega/\Gamma \approx 0.91$ for a three-level system and 1.23 ± 0.10 measured [5]. The damped circuit falls between the two, with its normal modes already split but the EIT model still preferred, and the undamped circuit is well beyond the switch. Damping at fixed coupling therefore moves the circuit across the EIT to ATS transition.

C. Crossover displacement

The measured crossings correspond to C_1 of 135 to 143 nF, not 104.9 nF, so $h_L = h_R$ does not locate the symmetric configuration. An error in the detuning axis cannot account for this, as it would require f_1 or f_2 to be about 7% from the values used, while the window tracks f_1 to within 2.5% in 19 of 20 runs and the reference peak agrees with f_2 to 0.3%.

The constant-resistance model crosses on the positive side, at +0.28 and +0.07 kHz with the DC values (damped and undamped) and at +0.59 and +0.24 kHz with the fitted values. Unequal constant damping therefore pushes the crossing the wrong way. The measured a lies above the constant-fit model in 13 of 13 damped runs, by +0.075 on average, and in 6 of 7 undamped runs. Since the heights of the peaks are set by the losses (Sec. II A), a loss rising with frequency would lower the high-frequency peak and move the zero to negative detuning. This is left as a hypothesis for future work.

D. Excess loss and open points

In the damped regime the excess over DC is about 8.7 Ω in R_1, about 87% of the winding resistance, and about 23 Ω in R_2, with 22.3 Ω from the reference run alone. The skin depth in copper at 20 kHz is 0.46 mm, which raises the resistance by under 2% for wire thinner than about 0.9 mm, so it cannot explain the excess. Proximity effect between turns or core loss could.

The transparency offsets remain unexplained. The models put f_T 0.1 to 1.0% below f_1, while the measured window is 1.4 to 2.4% below in the damped regime and 1.9 to 2.5% above in the undamped regime. An error in L_1 or C_1 would shift both regimes the same way. The undamped currents are about 20 times smaller, with a peak V_{L2} of 0.31 to 0.40 V against about 8 V in the damped regime. The offsets are reported as measured, with no claimed explanation.

VI. CONCLUSION

Sweeping a single capacitor through the symmetric configuration of the classical EIT circuit traverses one continuous family of lineshapes in which the sign of the peak asymmetry is set by the detuning. Mapped onto the atomic Λ system, the capacitor is a pure control detuning, the peaks follow the dressed states with no fitted parameter and the window stays at two-photon resonance. Damping at fixed coupling trades asymmetry for Fano character, with a slope ratio of 3.9 ± 0.7 between the two regimes, and the Akaike test of Refs. [4, 5] places the damped circuit on the EIT side and the undamped circuit on the ATS side of the transition.

The zero of the asymmetry is displaced to negative detuning, -1.45 ± 0.29 and -1.23 ± 0.15 kHz, in both regimes, and constant unequal damping does not explain it. The losses at the operating frequency exceed the DC values well beyond the skin effect, by about 87% in the dark mesh, and the opposite transparency offsets of the two regimes remain open. Future work should measure the resistances at the operating frequency and map the crossover against the damping.

ACKNOWLEDGMENTS

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[7] Reactance is the opposition a capacitor or inductor offers to alternating current. Like resistance it is measured in ohms and it limits current through $I = V/|Z|$, but unlike resistance it stores energy rather than dissipating it, and it depends on frequency. An inductor's reactance $X_L = \omega L$ rises with frequency; a capacitor's $X_C = 1/(\omega C)$ falls with it. In a series loop the two act in opposition, so the net reactance is $X = \omega L - 1/(\omega C)$: negative below the loop's resonance, zero at it, positive above it. The sign of X

therefore records which side of resonance the loop is on, and its magnitude how far.

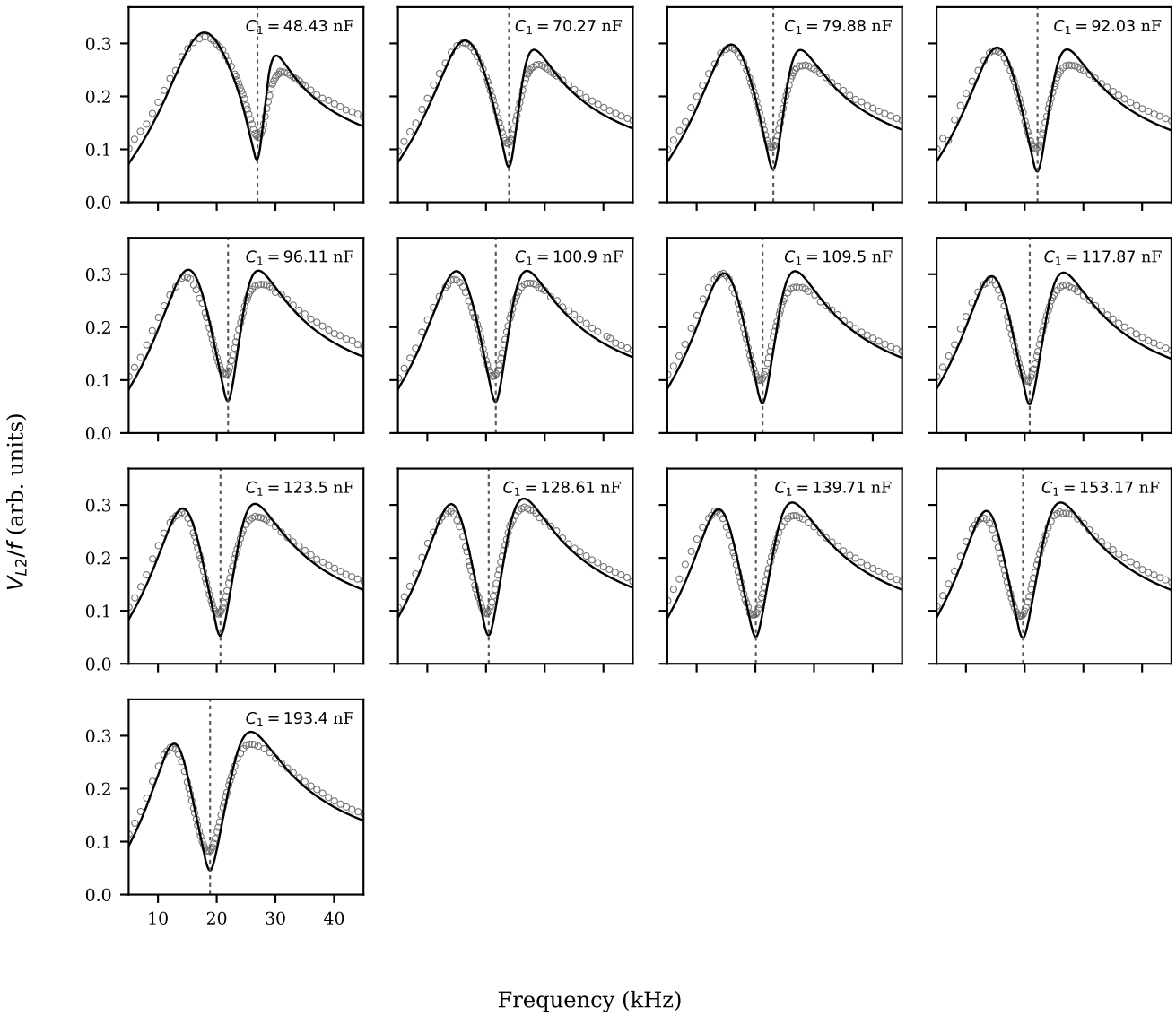

$C_1 = 48.43$ nF
$C_1 = 70.27$ nF
$C_1 = 79.88$ nF
$C_1 = 92.03$ nF
$C_1 = 96.11$ nF
$C_1 = 100.9$ nF
$C_1 = 109.5$ nF
$C_1 = 117.87$ nF
$C_1 = 123.5$ nF
$C_1 = 128.61$ nF
$C_1 = 139.71$ nF
$C_1 = 153.17$ nF
$C_1 = 193.4$ nF
V_{L2}/f (arb. units)
0.0
0.1
0.2
0.3
10
20
30
40
Frequency (kHz)

FIG. 3.

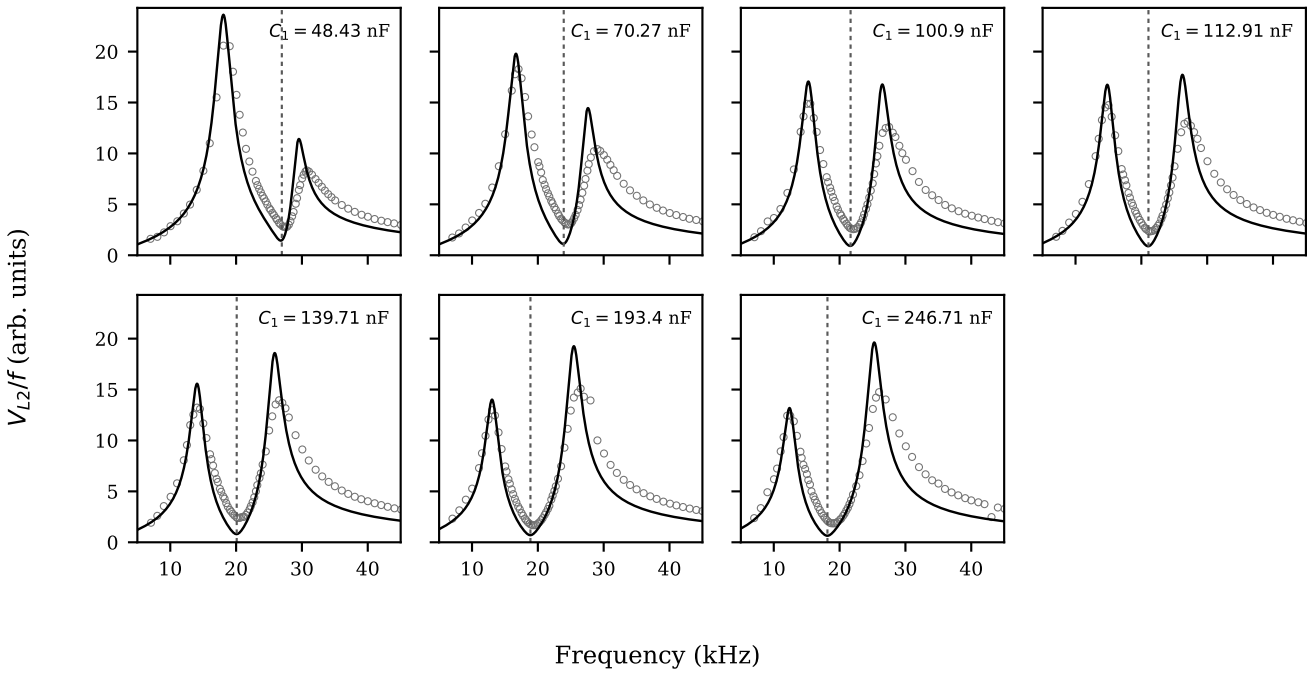

$C_1 = 48.43$ nF
$C_1 = 70.27$ nF
$C_1 = 100.9$ nF
$C_1 = 112.91$ nF
$C_1 = 139.71$ nF
$C_1 = 193.4$ nF
$C_1 = 246.71$ nF
V_{L2}/f (arb. units)
Frequency (kHz)

FIG. 4.

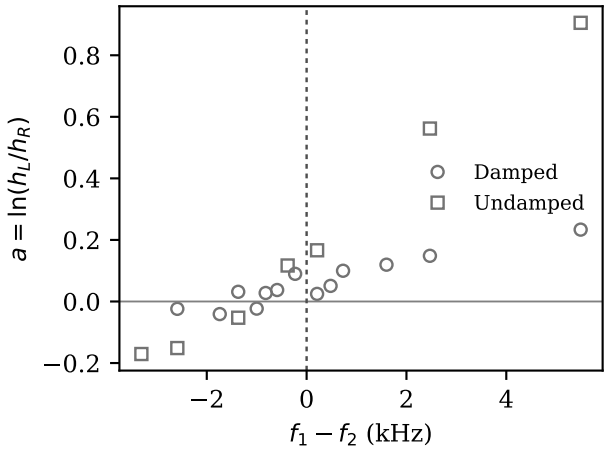

Damped
Undamped
$a = \ln(h_L/h_R)$
$f_1 - f_2$ (kHz)

FIG. 5.

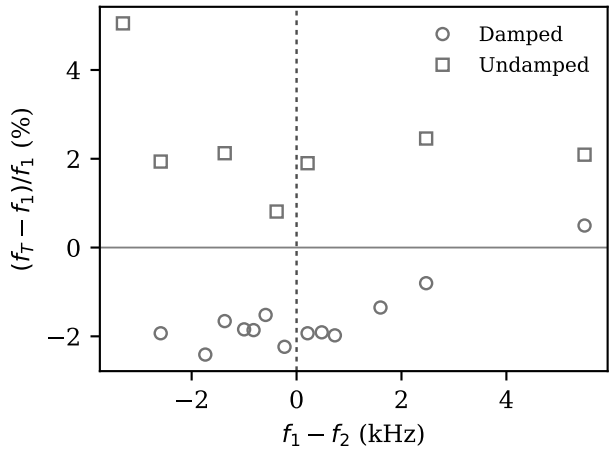

Damped
Undamped
$(f_T - f_1)/f_1$ (%)
$f_1 - f_2$ (kHz)

FIG. 6.

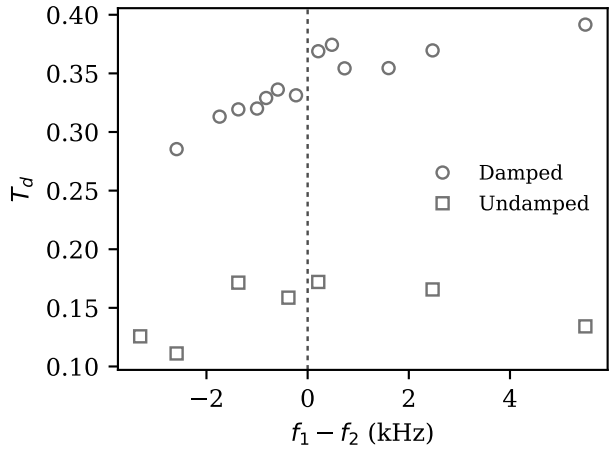

Damped
Undamped
T_d
$f_1 - f_2$ (kHz)

FIG. 7.

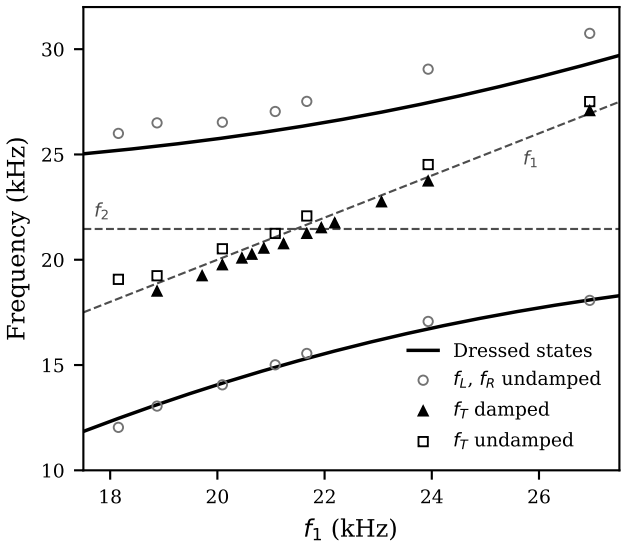


FIG. 8. Peak and window frequencies against the dark mesh resonance f_1, set by C_1. Lines: dressed states of Eq. (4), no fitted parameters, and the uncoupled resonances f_1 and f_2. Points: measured undamped f_L and f_R, and measured f_T in both regimes.

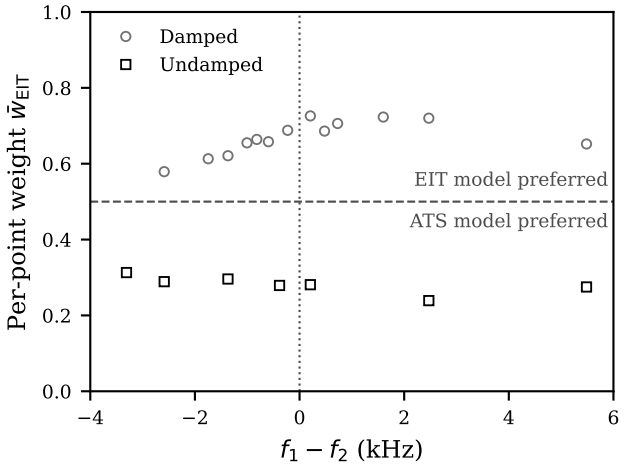


FIG. 9. Per-point Akaike weight of the EIT model, Table VIII, against the detuning $f_1 - f_2$ in both regimes. A weight of 0.5 means no preference between the models.